

BAHAA ALDEEN MOHAMED

Senior .NET Engineer — Solution Architecture & Systems Analysis

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PROFESSIONAL SUMMARY

Software architect and systems analyst working in the .NET ecosystem, with a track record of taking architectural decisions on systems that were considered too risky to change. Led the re-architecture of a distributed orchestrator to remove its transaction-coordinator dependency — the constraint that had frozen an entire service estate on unsupported runtimes — then authored the transformation template, its governing rules, and the AI-guided workflow that let other teams repeat it. A single upgrade cycle fell from 5-7 working days to a maximum of 3, with a company award for the time and cost saved. Designs and documents systems as well as building them: solution architecture documents, SRS and low-level design, and Mermaid workflow modelling. Architected payment middleware serving 9 countries on Clean Architecture with CQRS, Strategy, and Pipeline. Separately awarded as a subject matter expert for AI enablement across non-technical departments. Completing diplomas in microservices architecture and Azure, and targeting systems analysis and architecture roles.

TECHNICAL SKILLS

Architecture & Patterns: Clean Architecture, CQRS, Mediator, Repository, Strategy, Pipeline, Microservices, Distributed Transactions / DTC removal, Workflow Orchestration, OOP, SOLID

Systems Analysis & Design: Solution architecture documents, Software Requirements Specification (SRS), Low-Level Design (LLD), Architecture diagrams, Workflow and sequence modelling, Mermaid diagramming, Requirements analysis, Cross-team technical facilitation

Legacy Modernization: Framework upgrade strategy, Incremental migration, Project templating & rule authoring, Refactoring at scale, Cross-team transformation leadership

AI Engineering & Enablement: Microsoft Copilot Studio, Autonomous agent development, AI-guided development workflows, Claude Code, GitHub Copilot, AI training & curriculum design

Languages & Frameworks: C#, .NET 9 / .NET 10, .NET Core, .NET Framework, ASP.NET Core, ASP.NET MVC, Web API, Entity Framework Core, Dapper, Blazor

Data & Messaging: Microsoft SQL Server, LINQ, Query optimization, Redis, RabbitMQ

Cloud & DevOps: Microsoft Azure, Azure Static Web Apps, Azure DevOps, Team Foundation Server, Git, CI/CD pipelines

APIs & Integration: RESTful APIs, Payment gateway integration, JWT authentication, Google Maps API, reCAPTCHA, SMS & email gateways

Frontend: Angular, React, Blazor, HTML5, CSS3, JavaScript, jQuery, Ajax

Practice & Documentation: Agile / Scrum, API documentation, FluentValidation

PROFESSIONAL EXPERIENCE

Senior Full-Stack Software Engineer | Andalusia Business Solutions

November 2025 – Present | Alexandria, Egypt

- Led the re-architecture of the platform's Temporal orchestrator to remove its distributed transaction coordinator (DTC) dependency, unblocking the upgrade of legacy .NET Core 2.2 and .NET Framework 4.5 services.
- Authored the transformation project template and its governing rules as the first engineer on the initiative, giving every subsequent module upgrade a repeatable starting point.
- Designed a set of AI skills and an orchestrating workflow to guide the modernization, cutting the time to understand and upgrade one system cycle. Recognised with a company award for the resulting time and cost saving. **(5-7 working days reduced to a maximum of 3)**
- Built and delivered an AI training course for non-technical departments including HR, and was recognised as a subject matter expert for it.
- Developed an autonomous AI agent on Microsoft Copilot Studio.
- Modernized 5 platform modules, leading the transformation within the team and running cross-team sessions to guide other teams through the same migration.

Senior Full-Stack Software Engineer | PS Digital

November 2024 – November 2025 | Alexandria, Egypt

- Architected and built a middleware reverse proxy for payment gateway integration serving 9 countries, using Clean Architecture with CQRS, Strategy, and Pipeline patterns. **(9 countries)**
- Stepped in to cover the engineering manager's position during their sick leave, holding it for 45 days through a business-critical period, and was promoted on the strength of that period.
- Upgraded enterprise projects to current C# and .NET, introducing CQRS and the Mediator pattern to improve maintainability, testability, and separation of concerns.

Backend Software Engineer | MEEM Development

July 2024 – November 2024 | Egypt

- Produced solution architecture documentation — architecture diagrams and workflow designs — used by delivery teams as the reference for build work.
- Authored Software Requirements Specification (SRS) and Low-Level Design (LLD) documents, translating business requirements into unambiguous technical specifications that teams could implement without re-interpretation.

Software Engineer, promoted to Senior Software Engineer | PS Digital

March 2023 – July 2024 | Alexandria, Egypt

- Promoted twice within the first year on the strength of delivery.
- Led the rebuild of a legacy MVC web application into a modern .NET Web API, improving scalability and maintainability.
- Replaced Entity Framework with Dapper in the data access layer, improving query performance and simplifying data operations.

AWARDS AND RECOGNITION

- **Award for engineering time and cost saving** — Andalusia Business Solutions. For the AI-guided modernization workflow that cut a system upgrade cycle from 5-7 working days to a maximum of 3.
- **Subject Matter Expert recognition** — Andalusia Business Solutions. For designing and delivering an AI training course for non-technical departments, including HR.

EDUCATION

Microservices Architecture — Diploma (in progress) | Steps Learning Academy

January 2025 – Present

- Microservices design and Angular frontend development for scalable distributed systems.

Azure Cloud A-to-Z — Diploma (in progress) | Steps Learning Academy

January 2025 – Present

- Azure deployment, management, security, and cloud-native application practice.

Full Stack Development using .NET Framework — Diploma | Information Technology Institute (ITI)

November 2022 – March 2023

- Intensive 4-month bootcamp on the Microsoft stack, covering computer science fundamentals.
- C# and .NET: built web applications and APIs.
- SQL Server: relational database design and management.
- Angular: dynamic web interfaces.

Tourism Studies | Alexandria University

February 2014 – February 2019

CERTIFICATIONS

- IT Essentials — Cisco Networking Academy (January 2023)
- Software Architecture: From Developer to Architect — LinkedIn Learning (July 2024)
- Microservices Design Patterns — LinkedIn Learning (September 2024)
- Azure Microservices with .NET Core for Developers — LinkedIn Learning (August 2024)
- Learning RabbitMQ — LinkedIn Learning (August 2024)

12 more, plus case studies and architecture diagrams, at bahaa37.github.io.